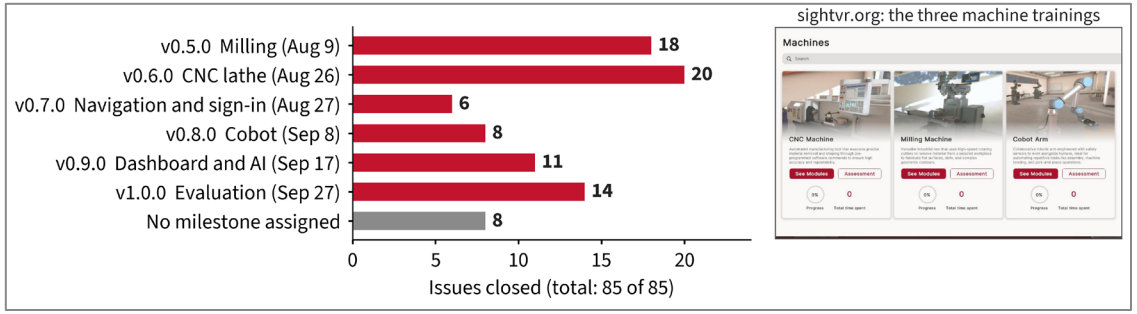
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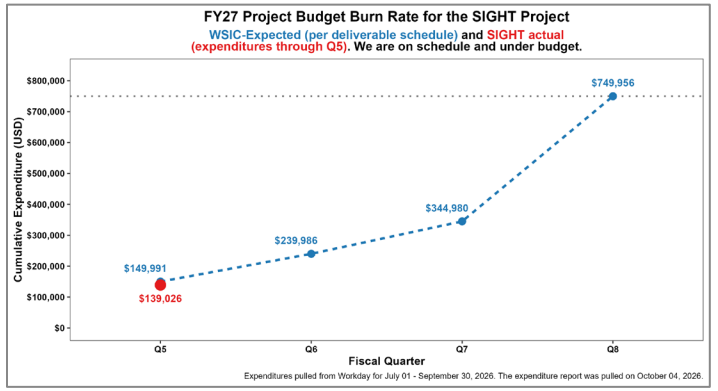
SIGHT: Quarter 5 Status

Track 2: Proof of Concept | **PI:** Fadel M. Megahed | **Status:** On Schedule and Under Budget

- **Completion:** >64.4% (11 of 15 tasks and 5 of 9 success metrics complete)
- **Budget:** $139,026 spent in Q5, against a WSIC-expected value of $149,991
- **Deliverables:** BWC accepted the Interim Progress Report and approved all 16 Year 1 deliverables
- **VR platform:** under Miami control since July 10, 2026, and public at sightvr.org
- **SIGHT-VR v1.0.0:** released on September 29, 2026, after eight releases that closed 85 of 85 issues
- **Team:** two new student researchers; Dr. Yousif is now an Associate Research Scientist



VR issues closed in Q5 by release, and the trainings at sightvr.org



FY27 expenditures (red) against the WSIC-expected value (blue)

Source: DO7.4 Quarterly Progress Report (Quarter 5, July 1 to September 30, 2026)

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Extension of the VR Training to Two Makino Machines

Task 5.3. This work is in addition to the three machines in our proposal.



Narrated video (1:47). Click the picture to play, or watch it on YouTube.

- Two new trainings, **six guided modules each**: the Makino D200Z (5-axis machining center) and the UX3i (wire EDM)
- D200Z training ready **overnight**; UX3i training in **approximately two days**
- First three machines: **119 to 172 days** each, with a vendor team and new 3D assets
- Made with our AI-based pipeline, on our own initiative
- The trainings include machine faults, a feature that visitors to our OSC 2026 poster asked for
- Next: a meeting with Makino's Corporate Training Manager

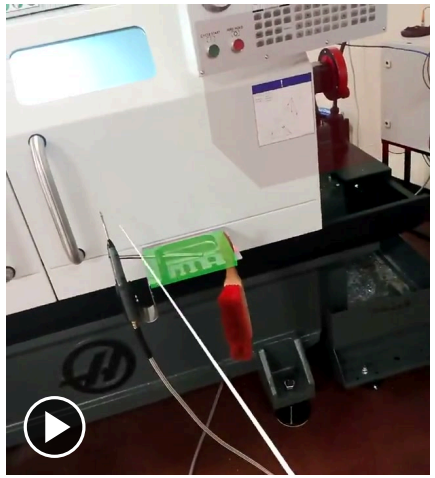
Source: DO7.4 Quarterly Progress Report (Quarter 5, July 1 to September 30, 2026)

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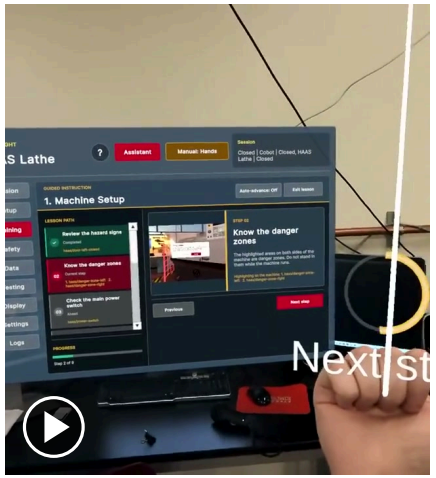
AR on the Magic Leap 2 Headset

Tasks 4.4 and 5.2. Developed by our Miami team.

- **Cobot instructional mode technically complete:** seven modules, each recorded on the headset (videos)
- **Live machine state** from the edge device, and **safety zones** that warn the user (4-ft boundary around the cobot)
- **Operational mode prototype, in addition to our proposal:** the headset sends commands to the cobot
- **Machine SOPs revised for AR;** first SOP-guided run of the Bridgeport mill on September 29, 2026



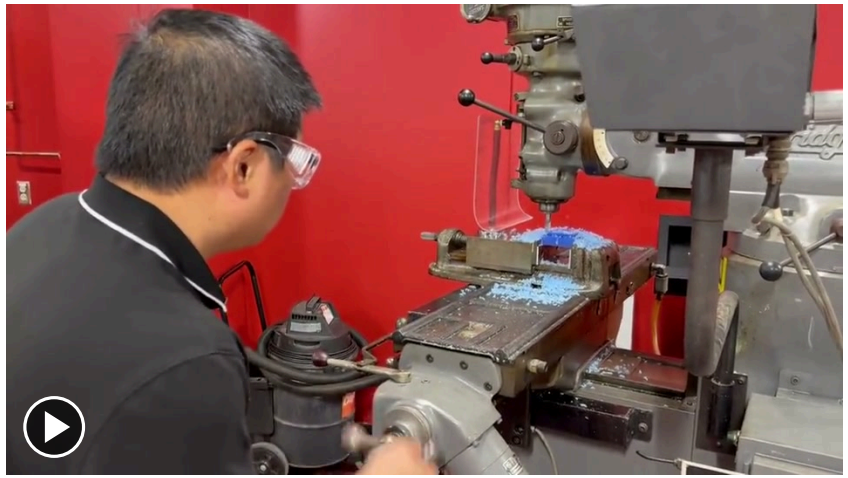
Cobot instructional module (0:21)



Hand gesture to change step (0:06)



Operational mode (video)



First SOP-guided mill run (0:15)

Source: DO7.4 Quarterly Progress Report (Quarter 5, July 1 to September 30, 2026)

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Pilot Test Plan, Team and Partners, and Quarter 6 Work

Pilot test (Task 5.1)

- Within-subject design: each participant trains on the three machines, in counterbalanced order
- **Recruit 42 participants**; at least 36 must complete (Metric 8)
- Measures: hazard quiz, technology acceptance survey, interview, observation sheet, and app logs
- **VR study first**, in person at the AM Hub; AR study follows
- **Tests: December 2026 to March 2027**

Team and partners

- **Dr. Cavuoto:** VR review (27 of the 85 issues) and the draft protocol
- **Makino:** meeting on the two VR trainings in preparation
- **Industrious Group:** site visit on September 25, 2026
- **Yamaha:** feedback on sightvr.org from its team
- **MaxByte:** image annotation for AR
- **New student researchers:**



Wissam Nusair



Lindsay Stevenson

Quarter 6 work

- AR: instructional content for the Bridgeport mill; live digital twin in the headset
- VR: start version 2.0, with assessments that differ from the training
- Pilot test: complete the protocol, submit the IRB amendment, run pre-pilot sessions, and start the tests in December
- Dissemination: OSC 2027 and ASSP Safety 2027 submissions under review

Source: DO7.4 Quarterly Progress Report (Quarter 5, July 1 to September 30, 2026)